FDF-ROCKET: A False Discovery Rate Filtering Approach for Explainable Time Series Classification

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Abstract

The Random Convolutional Kernel Transform (ROCKET) is a highly accurate and computationally efficient approach to time-series classification, but most existing explainability methods do not effectively exploit its unique structure. Because ROCKET relies on thousands of convolutional kernels with randomly generated weights, many of its derived features are unlikely to contain meaningful predictive information, potentially limiting the effectiveness of explainability approaches that treat all features equally. We propose a statistically motivated explainability method, False Discovery Filtered ROCKET (FDF-ROCKET), that frames the final classifier stage of ROCKET as a multiple-testing problem and uses empirical Bayes local false discovery rate estimates to filter out the effects of unimportant kernels when identifying influential temporal regions in the input data. We evaluate our method using simulation studies that introduce highly localized patterns at isolated time points and extended temporal regions. In both settings, we find FDF-ROCKET more reliably identifies influential points than competing approaches including layer-wise relevance propagation (ROCKET-LRP), as well as time-series adaptations of Shapley Additive Explanations (Kernel SHAP) and local interpretable model-agnostic explanations (LIME). We further demonstrate the utility of FDF-ROCKET on a real data case study involving the camera-based monitoring of pupil diameter in alcohol-impaired and sober drivers, showing FDF-ROCKET produces explanations with superior faithfulness and unique insights into class differences. Together, our findings highlight the potential advantages of integrating false discovery rate inference into explainability methods for time-series classification.

Keywords: time-series classification, statistical learning, false discovery rate, model interpretability

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1 Introduction

Time-series classification, or the use of statistical and machine learning methods to categorize time-ordered sequential data with discrete class labels, is becoming increasingly prevalent due to advances in the ease at which streams of time-ordered data can be collected and stored. Existing applications include biomedicine [1], finance [2], transportation safety [3], and many other domains [4]. Advances in convolutional neural networks (CNNs) have been an important development in time-series classification, with many of the top modeling approaches relying on convolution operations to learn predictive temporal trends [5]. Among the top performing methods, the Random Convolutional Kernel Transform (ROCKET) [6] is unique. Rather than using a variation of gradient descent to learn data-driven values for the weight parameters in the model's convolution kernels, ROCKET uses a large collection of kernels with random weights to generate features used in a final classifier, thereby making it substantially more computationally efficient than CNNs.

While convolution-based machine learning methods offer high predictive accuracy, many applications demand both predictive accuracy and explainability. For tabular data, top performing machine learning approaches like gradient boosting (XGBoost) and random forests have a degree of inherent interpretability via gain-based importance measures. Additionally, methods such as Shapley Additive Values (SHAP) [7], Local Interpretable Model-agnostic Explanations (LIME) [8] can be used for models lacking intrinsic interpretability. However, for state-of-the-art time-series classification models, explainability becomes more complicated. Many explainability methods for time-series classification, including Cross-Domain Integrated Gradients (CDIG) [9] and Perturbation by Prioritized Replacement (PERT) [10], have been developed for neural-network architectures and require access to model gradients. This creates challenges for ROCKET, which does not involve gradients, thereby necessitating different approaches to explainability.

Layer-wise Relevance Propagation (ROCKET-LRP) has been developed as a model-intrinsic local explanation technique for ROCKET classifiers [11]. Meanwhile, kernel SHAP, a computationally efficient extension of SHAP relying on statistical sampling rather than all possible coalitions of features, can be used on time-series classification models [7], including ROCKET classifiers, and LIME has also been adapted to time-series classification [8]. However, because kernel SHAP and LIME are model-agnostic, it is unclear how well they will handle the unique nature of ROCKET, where thousands of randomly generated convolutional kernels with random weights are used to construct features, many of which are not expected to contain meaningful predictive information. Furthermore, even existing model-specific approaches such as ROCKET-LRP do not explicitly account for the fact that many of ROCKET's kernels are likely non-informative.

Because ROCKET relies on thousands of kernel-derived features, distinguishing informative kernels from noise naturally resembles a large-scale multiple-testing problem. Owing to this unique structure, we propose a false discovery

rate based approach to the explainability of ROCKET classifiers, a method we refer to as False Discovery Filtered ROCKET, or FDF-ROCKET. Our approach can be used to identify influential temporal regions, thereby allowing analysts to better understand predictive temporal patterns. Existing approaches either propagate relevance over all kernels, or do not capitalize on ROCKET's unique structure, potentially leading to less effective identification of relevant time points. FDF-ROCKET uses empirical Bayes methods to estimate the posterior probability that each kernel-derived feature contributes meaningfully to classification, and restricts its explanations to use only statistically supported kernels. We show using simulation studies and a real-world case study involving pupil diameter time-series data captured by a driver monitoring system in a study of alcohol-impaired driving, that our approach more effectively identifies influential time points in comparison with ROCKET-LRP, kernel SHAP, and LIME.

2 Methods

2.1 Random Convolutional Kernel Transform (ROCKET)

ROCKET classifiers have achieved high levels of classification performance on benchmark time-series classification datasets with significantly lower computational complexity and runtime than CNNs and other related deep learning models [6, 5]. ROCKET uses a large number of convolutional kernels, indexed by $k \in \{1, \ldots, K\}$, each with random weights, $\mathbf{w}_k$, with varying lengths, l_k, and dilation, d_k. For each observed sample, indexed by $i \in \{1, \ldots, n\}$, each kernel is convolved with the time-series input, $\mathbf{X}_i$, a p by t matrix of input channels and time steps. Zero-padding is typically applied to $\mathbf{X}_i$ to ensure all parts of each kernel are exposed to the input, and a bias, b_k, whose value is also determined at random is added to convolutional output. Thus, the application of the kth kernel to the ith subject produces the output $\tilde{\mathbf{X}}_{ik}$, at the jth time step defined by:

$$\tilde{\mathbf{X}}_{ik}[j] = \sum_{c=1}^{p} \sum_{m=0}^{l_k - 1} \mathbf{w}_k[c, m] \cdot \mathbf{X}_i[c, j + m \cdot d_k] + b_k \tag{1}$$

The output of (1) is not directly used by the final classifier, but instead two derived features are used, the proportion of positive values (PPV), and the global pooled maximum (MAX), respectively given by:

$$\text{PPV}_{ik} = \frac{1}{J_k} \sum_{j=1}^{J_k} \mathbb{1}\left(\tilde{\mathbf{X}}_{ik}[j] > 0\right)$$

and

$$\text{MAX}_{ik} = \max_{j \in \{1, \ldots, J_k\}} \left(\tilde{\mathbf{X}}_{ik}[j]\right)$$

for the ith sample, where J_k is the number of steps in the convolution output for the kth kernel.

MAX and PPV remove the time-ordering of the data, thereby allowing the ROCKET transformed data to be represented using an n by $2K$ matrix, which we denote $\mathbf{Z}$. A final classifier, $f(\cdot)$, is then used to determine predicted probabilities for each sample such that $\hat{\pi} = f(\mathbf{Z})$, in the case of binary classification, which in turn can be mapped to predicted categories, $\hat{\mathbf{y}}$. Because thousands of randomly generated kernels were used to create the collection of MAX and PPV features used in the final classifier, only a subset should be expected to meaningfully contribute to predictive performance, thereby motivating the treatment of kernel-derived features as candidates in a multiple-testing framework.

2.2 FDF-ROCKET

Our proposed explanation method, FDF-ROCKET, involves the following steps:

- Fitting an appropriate final classifier to the MAX and PPV features.
- Finding a suitable collection of feature-specific test statistics using the final classifier.
- Applying a local false discovery rate filtering criteria to these statistics in order to identify a set of kernels whose derived features have sufficient statistical support.
- Activation propagation using the selected kernels.
- Aggregation and thresholding of the activated series to identify influential time points.

The following subsections describe each component of the FDF-ROCKET procedure in detail.

2.2.1 Final Classifier

The first step of FDF-ROCKET is to fit a classification model using the ROCKET-derived MAX and PPV features. While any classifier could be used here, we focus on L1-penalized regression due to its ability to induce sparsity in high-dimensional settings. This is particularly appropriate for ROCKET, where many kernel-derived features are expected to be uninformative.

To begin, let $l(\beta)$ denote the logistic regression log-likelihood function, where β is a vector of length $2K$ of the coefficients in the final classifier. Additionally, let $\mathbf{Z}$ denote the transformed collection of MAX and PPV, and $\mathbf{y}$ is a vector of length n containing binary outcomes. While we focus on binary classification, we note that our approach could be generalized to multi-class applications using the multinomial likelihood.

The coefficient vector, β, contains important information about the relative contributions of each MAX and PPV, and thus the importance of their corresponding kernels. β is found by minimizing the objective function:

$$Q(\beta) = -l(\beta) + \lambda||\beta||_1$$

where λ is the regularization parameter.

This minimization has no closed form solution and requires an iterative approach. One such approach is to approximate the log-likelihood using a second-order Taylor series about $\tilde{\beta}$, a working estimate of the coefficient vector:

$$l(\beta) \approx l(\tilde{\beta}) + (\beta - \tilde{\beta})^T \nabla l(\tilde{\beta}) + \frac{1}{2}(\beta - \tilde{\beta})^T \nabla^2 l(\tilde{\beta})(\beta - \tilde{\beta})$$

Letting $\eta = \mathbf{Z}\beta$ and $\tilde{\eta} = \mathbf{Z}\tilde{\beta}$, the Taylor series approximation can be factored into a quadratic expression:

$$\frac{1}{2}(\tilde{\mathbf{y}} - \eta)^T \mathbf{W}(\tilde{\mathbf{y}} - \eta) + \text{const.}$$

where $\tilde{\mathbf{y}} = \tilde{\eta} - \nabla^2 l(\tilde{\eta})^{-1} \nabla l(\tilde{\eta})$ and $\mathbf{W} = -\nabla^2 l(\tilde{\boldsymbol{\eta}})$.

This allows an iteratively reweighted least squares (IRLS) algorithm using $\tilde{\mathbf{y}}$ as a pseudo-response to find the estimates, $\hat{\beta}$, which minimize $Q(\cdot)$.

2.2.2 Feature-Specific Test Statistics

The optimization of $Q(\beta)$ is relevant because it provides the basis for the construction of feature-specific test statistics that underlie our false discovery rate filtering approach. The solution to the L1-penalized IRLS problem described in the previous section must satisfy the Karush-Kuhn-Tucker (KKT) optimality conditions [12], which mathematically characterize when a ROCKET-derived feature makes a non-zero contribution to the final classifier. More specifically, the KKT conditions imply that the vth feature from $\mathbf{Z}$ will only have a non-zero effect if:

$$\tfrac{1}{n}|\mathbf{z}_v^T \mathbf{W}(\hat{\mathbf{y}} - \mathbf{Z}_{-v}\hat{\beta}_{-v})| > \lambda \tag{2}$$

where the subscript $-v$ denotes the removal of the vth feature.

The quantity $\mathbf{z}_v^T \mathbf{W}(\hat{\mathbf{y}} - \mathbf{Z}_{-v}\hat{\boldsymbol{\beta}}_{-v})$ in Equation (2) can be seen as a weighted score measuring the association of the vth feature and the pseudo-residuals that persists after accounting for the other features used by the classifier. Using this quantity, Miller and Breheny [13, 14] showed that under the appropriate regularity conditions the standardized statistic:

$$T_v = \frac{\mathbf{z}_v^T \mathbf{W}(\hat{\mathbf{y}} - \mathbf{Z}_{-v}\hat{\beta}_{-v})}{\sqrt{n\mathbf{z}_v^T \mathbf{W}\mathbf{z}_v}} \tag{3}$$

is asymptotically Normal when the vth feature is marginally independent of the response.

Thus, the collection of T_v for all $v \in \{1, \ldots, 2K\}$ provides a reasonable basis for simultaneously evaluating the statistical support for each ROCKET-derived feature. Unlike most other multiple-testing approaches, this support is assessed in the context of the final classifier rather than using many univariate hypothesis tests.

Furthermore, the innate characteristics of ROCKET make the assumptions used to derive T_v highly plausible in practice. In particular, the strongest condition requires weak dependence among the features unrelated to $\mathbf{y}$, or $\frac{1}{n}\mathbf{z}_v^T\mathbf{W}\mathbf{Z}_{-v} \xrightarrow{p} \mathbf{0}$ as n goes to infinity. For ROCKET this is reasonable because features are generated using randomly constructed kernels with differing weights, lengths, dilations, and biases, so features that do not encode meaningful predictive information are unlikely to have strong systematic dependencies.

2.2.3 False Discovery Rate Filtering

The collection of statistics, $\{T_1, \ldots, T_{2K}\}$, from (3) provides a measure of the evidence for the contribution of each MAX and PPV in the final classifier. In order to apply false discovery rate filtering using these statistics to identify convolutional kernels of interest, we follow the approach of Stephens [15], and assume the statistics arise from a two-component mixture model:

$$T_v \sim g(u) = \pi_0 f_0(u) + (1 - \pi_0) f_1(u)$$

where $f_0(\cdot)$ denotes the null density, $f_1(\cdot)$ denotes the non-null density, and π_0 is the proportion of null features.

Stephens' adaptive shrinkage (ash) framework estimates the marginal density $g(\cdot)$ from the observed test statistics using an empirical Bayes mixture model. The local false discovery rate for feature v is then given by:

$$\text{lfdr}_v = P(H_{0v} \mid T_v) = \frac{\pi_0 f_0(T_v)}{g(T_v)}, \tag{4}$$

which may be interpreted under the empirical Bayes framework as the posterior probability that the vth feature is non-influential given the information encoded in its test statistic [16]. Thus, features with smaller local false discovery rates have stronger statistical evidence supporting their relevance in the final classifier.

Because FDF-ROCKET aims to only consider kernels with statistically meaningful evidence of contribution to predictive performance, kernel k is retained if at least one of its associated features satisfies:

$$S = \left\{ k : \text{lfdr}_k^{\text{PPV}} \leq \alpha \text{ or } \text{lfdr}_k^{\text{MAX}} \leq \alpha \right\} \tag{5}$$

where S denotes the set of selected kernels, and α is a user-specified threshold imposed on the local false discovery rates from (4).

This filtering step distinguishes FDF-ROCKET from other ROCKET-specific explainability approaches by explicitly considering the statistical evidence in support of the features produced by each kernel. The value of α can be tuned

to control the trade-off between sensitivity and specificity, with smaller values imposing stricter requirements and leading to less complex annotations. The simulation studies presented in Section 3 each consider several choices of α.

2.2.4 Activation Propagation

After kernels have been identified using false discovery rate filtering, the contributions of the selected kernels must be mapped back to the original time domain. We achieve this by applying only the selected kernels, then summing the absolute values of the convolution outputs along aligned time-steps. Thus, the activated output for the ith sample at the jth time step is given by:

$$A_i[j] = \sum_{k=1}^{K} \mathbb{1}(k \in S) \sum_{c=1}^{p} \left| \tilde{\mathbf{X}}_{ik}[c, j] \right| \tag{6}$$

where p is the number of input channels and S indexes the set of selected kernels from (5).

The aggregation performed in (6) produces an output whose length aligns with the original time-series, with larger values reflecting stronger cumulative evidence that the observed data at a particular time-step contributed to the classifier's prediction. Because the activated output in (6) is not inherently sparse, an additional thresholding step must be applied to arrive at a final set of influential time-steps. We recommend either the top M largest values, or all values which are a set number of standard deviations larger than zero (such as 2 or 3).

3 Synthetic Data Experiments

To evaluate the effectiveness of FDF-ROCKET, we consider a set of synthetic data simulations similar to those conducted by Ling et al. [11] in their evaluation of ROCKET-LRP. Each of these two experiments introduces a known trend into a randomly chosen subset of simulated univariate time-series such that the identity of all influential time-steps is known.

Unperturbed time-series data are generated using an order-2 normalized moving average process (NARMA2), and all experiments use ROCKET models that are trained on $n = 1400$ samples of length $t = 60$, with half of the samples containing an injected trend. All methods are evaluated on an independently generated set of $n = 300$ samples containing the injected trend. We report results across 100 simulation repetitions, each using a different randomization seed, for FDF-ROCKET, ROCKET-LRP, Kernel SHAP, and LIME. We also include the random selection of influential time points to provide a baseline illustrating performance levels corresponding with random chance in each experiment.

3.1 Spike Detection

Our first experiment adds a random number of spikes of magnitude 0.2 to the samples selected to contain an injected trend. The number of spikes that are injected is randomly drawn from a Poisson distribution with a mean of 2, and spikes are randomly inserted at individual time points within the sample.

Each method is used to flag only the number of time points corresponding with the true number of spikes that are present in an evaluation sample. To allow leniency, we consider a true positive identification to occur whenever a method flags a time point within ± 1 time-steps of the actual location of the spike. All other flagged locations are considered false positives.

The major challenge introduced in the spike experiment is that the true signal is localized to a very small number of isolated time points. So, successful explanations must be able to identify abrupt events while avoiding false positive identifications elsewhere in the series.

Method	Mean	90%	75%	50%	25%	0%
FDF-ROCKET (0.001)	**0.437**	**0.541**	**0.523**	**0.442**	0.393	0.186
FDF-ROCKET (0.01)	0.431	0.529	0.461	0.430	**0.405**	0.199
FDF-ROCKET (0.1)	0.379	0.471	0.442	0.390	0.326	0.153
Kernel SHAP	0.367	0.394	0.383	0.367	0.353	**0.317**
ROCKET-LRP	0.155	0.175	0.162	0.153	0.146	0.119
Random	0.139	0.158	0.149	0.140	0.129	0.104
LIME	0.133	0.150	0.142	0.133	0.125	0.089

Table 1: Distribution of recall by method for the spike experiment across 100 simulation repetitions. The best performance approach at each percentile is highlighted in bold.

For FDF-ROCKET, we report results for three different local false discovery rate thresholds, $\alpha \in \{0.1, 0.01, 0.001\}$. Table 1 summarizes the distribution of recall (true positive rate) across the 100 simulation iterations. In general, FDF-ROCKET is most effective at detecting the locations of inserted spikes, with the thresholds on the false discovery rate filter corresponding with high recall, particularly at the upper end of the distribution. Kernel SHAP on average performs similar to FDF-ROCKET with a lax false discovery rate threshold of 0.1, and exhibits the most consistent performance across simulation runs, showing the best performance in its worst simulation iteration. Both ROCKET-LRP and LIME struggled to identify spikes beyond what could be expected by random chance.

Pairwise statistical comparisons of each approach using Wilcoxon signed-rank tests of the mean recall achieved by the method across the 100 simulation repetitions are summarized in Figure 1. FDF-ROCKET using $\alpha = 0.01$ and $\alpha = 0.001$ exhibit significantly better recall than competing methods, but do not differ from each other. Kernel SHAP is the only competing method that performs better than randomly guessing where the injected spikes are located.

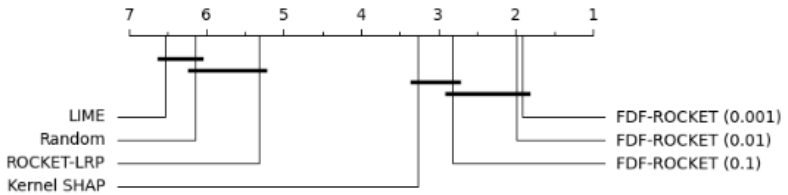


Figure 1: Critical difference diagram comparing the performance of competing methods in the spike experiment. Approaches connected by a bold black line do not exhibit a statistically significant difference in performance across 100 simulation iterations.

3.2 Shapelet Detection

In our second experiment we insert a sinusoidal wave spanning 10 time steps with some Gaussian noise into a randomly selected position in the previously described NARMA(2) time-series. Each method is used to flag its 10 most influential time steps, and we consider all points where the wave was inserted to be true positives with no allowed margin of error.

Unlike the spike experiment, successful shapelet detection involves the identification of an extended temporal region rather than a collection of isolated points. Thus, this experiment assesses each method's ability to identify sustained changes rather than abrupt events.

Method	Mean	90%	75%	Median	25%	Minimum
FDF-ROCKET	**0.776**	**0.803**	**0.800**	**0.795**	**0.784**	**0.691**
Kernel SHAP	0.539	0.553	0.544	0.539	0.532	0.514
ROCKET-LRP	0.245	0.260	0.252	0.244	0.239	0.225
Random	0.163	0.167	0.166	0.165	0.163	0.120
LIME	0.106	0.118	0.113	0.105	0.099	0.088

Table 2: Distribution of recall by method for the shapelet experiment across 100 simulation repetitions. The best performance approach at each percentile is highlighted in bold. FDF-ROCKET results are shown in a single row due to each α threshold selecting an identical set of kernels.

Table 2 summarizes the distribution of recall for the shapelet experiment across 100 simulation iterations. FDF-ROCKET is most effective at accurately identifying time steps that are part of the inserted sinusoidal wave. The relative ordering of methods is the same as it was in the spike experiment, though both Kernel SHAP and ROCKET-LRP perform appreciably better than random guessing.

The critical difference diagram displayed in Figure 2 further illustrates the superior performance of FDF-ROCKET in the shapelet experiment, with the difference between FDF-ROCKET and all competing methods being statistically significant. Both Kernel SHAP and ROCKET-LRP are significantly bet-

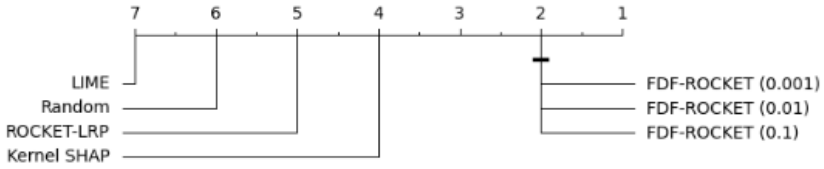


Figure 2: Critical Difference Diagram for Shapelet Experiment Recalls

ter than random guessing, and different from each other with Kernel SHAP performing better.

4 Alcohol-Impaired Driving Application

We next evaluate FDF-ROCKET using real-world ocular data from an alcohol-impaired driving study. In the study, $n = 26$ subjects completed a baseline (sober) drive in an advanced driving simulator, followed by alcohol dosing and a comparable follow-up drive. The simulator was equipped with a camera-based driver monitoring system that captured a 60Hz time-series of various ocular features. We focus on the mean pupil diameter, averaged over the left and right eye measurements, of each subject during an approximately 8-minute stretch of straight, rural driving in their baseline and post-dose drives, aiming to develop and understand a model that accurately differentiates between instances of sober and alcohol-impaired driving.

Each drive was divided into 60-second (3600 time-step) data samples. All samples from a given subject were assigned to either a training or validation set and ROCKET models using L1-penalized logistic regression as the final classifier were fit and evaluated for 10 different training-validation splits, each randomly assigning 21 (80.8%) of subjects to the training set and withholding all samples from the remaining 5 (19.2%) subjects for validation. In our results we report the mean area under the receiver operating characteristic curve (AUC) across these 10 validation sets.

4.1 Faithfulness

Faithfulness assesses the extent to which a set of features/time-steps identified by an explainability method are genuinely used by a predictive model by masking or perturbing the identified features/time-steps and measuring the resulting change in performance [17, 8]. Analyzing faithfulness is important in this application because the ground-truth status of each time point is unknown, so it offers us an objective, albeit indirect, measure of a time-step's importance. We apply FDF-ROCKET, ROCKET-LRP, kernel SHAP, and LIME to each of our models and mask the top 5%, 10%, and 15% of points identified by each method using sample-wise means.

Without any masking, our approach achieves a baseline AUC of 0.753 across validation sets. Table 3 shows the changes in classification performance after masking the top 5%, 10%, and 15% of most influential time-steps, with larger decreases in performance reflecting greater importance of the masked features.

Method	**5% ↓**	**10% ↓**	**15% ↓**
FDF-ROCKET (0.1)	**0.658 (0.083)**	**0.583 (0.089)**	**0.523 (0.075)**
FDF-ROCKET (0.01)	0.693 (0.059)	0.600 (0.094)	0.530 (0.086)
FDF-ROCKET (0.001)	0.695 (0.060)	0.620 (0.073)	0.557 (0.087)
ROCKET-LRP	0.682 (0.060)	0.655 (0.054)	0.625 (0.057)
Kernel SHAP	0.752 (0.046)	0.741 (0.051)	0.731 (0.052)
LIME	0.746 (0.048)	0.747 (0.047)	0.740 (0.053)
Random	0.746 (0.048)	0.745 (0.049)	0.740 (0.053)

Table 3: Mean (standard deviation) AUC across held out validation sets after masking the top 5%, 10%, and 15% most relevant time points.

FDF-ROCKET using $\alpha = 0.1$ exhibits the greatest decreases in AUC, suggesting it is most effective in identifying influential time-points. ROCKET-LRP performs slightly better than FDF-ROCKET when smaller values of α are used, but only looking at the top 5% of identified points. When considering larger amounts of time-points, FDF-ROCKET tends to identify points with greater predictive value than ROCKET-LRP, as evidenced by greater declines in AUC when using the top 10% and 15% of points. Kernel SHAP and LIME perform similar to randomly assigning masks, suggesting they were not effectively identifying influential time-points in this application.

4.2 Explanatory Agreement

Our analysis of faithfulness finds that both FDF-ROCKET and ROCKET-LRP are capable of identifying influential time-points, but the extent to which both methods are identifying the same points is unclear. Table 4 summarizes the correspondence between the top 5% of time-points flagged as influential by FDF-ROCKET (using $\alpha = 0.1$) and ROCKET-LRP across all validation sets.

	FDF Flagged	FDF Not Flagged
LRP Flagged	3863	18277
LRP Not Flagged	18277	402383

Table 4: Correspondence between FDF-ROCKET ($\alpha = 0.1$) and ROCKET-LRP time points identified as influential across test sets.

Despite having similar objectives, the two methods tend to identify different points, agreeing on only 3863 of 22140 (17.4%) selections. Follow-up analysis suggests that ROCKET-LRP identifies comparatively more blinks, which correspond with mean pupil diameters of exactly zero, and comparatively fewer points with values on the lower end of the pupil diameter distribution, with Figure 3

displaying the distribution of mean pupil diameter for the points selected by each method. This distinction is potentially important because alcohol-induced changes in ocular behavior are unlikely to be explained by differences in blink behavior alone, so FDF-ROCKET seems to be more sensitive to detecting more subtle changes in pupil dynamics.

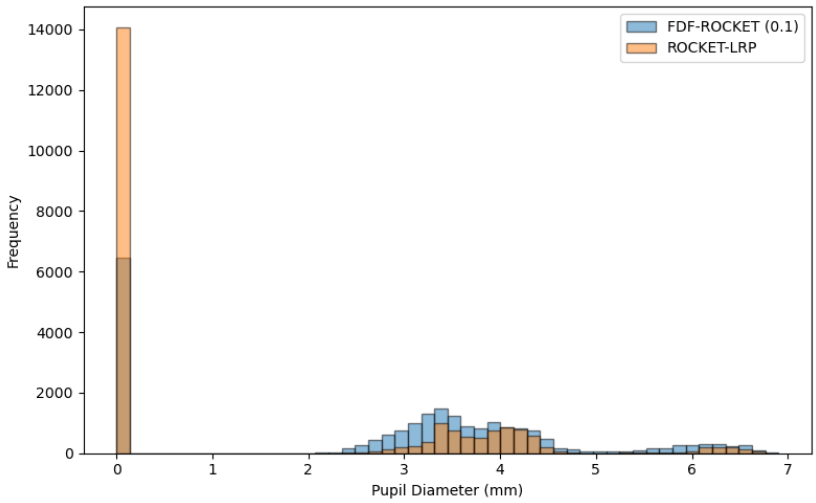


Figure 3: Distribution of the Pupil Diameter values (in mm) for the Top 5% Relevant points across test sets using ROCKET-LRP and FDF-ROCKET with an $\alpha = 0.1$ false discovery rate filter.

Differences between FDF-ROCKET and ROCKET-LRP can be further assessed visually for individual validation samples, with Figure 4 highlighting the top 5% of selected points for one of Subject 6's samples. We again see that ROCKET-LRP tends to concentrate importance on blinks, while FDF-ROCKET highlights additional periods of deviation from normal pupil behavior. When combined with our faithfulness results, FDF-ROCKET can be seen as more effectively picking up on slight changes in the pupils that are not driven by changes in blink behavior.

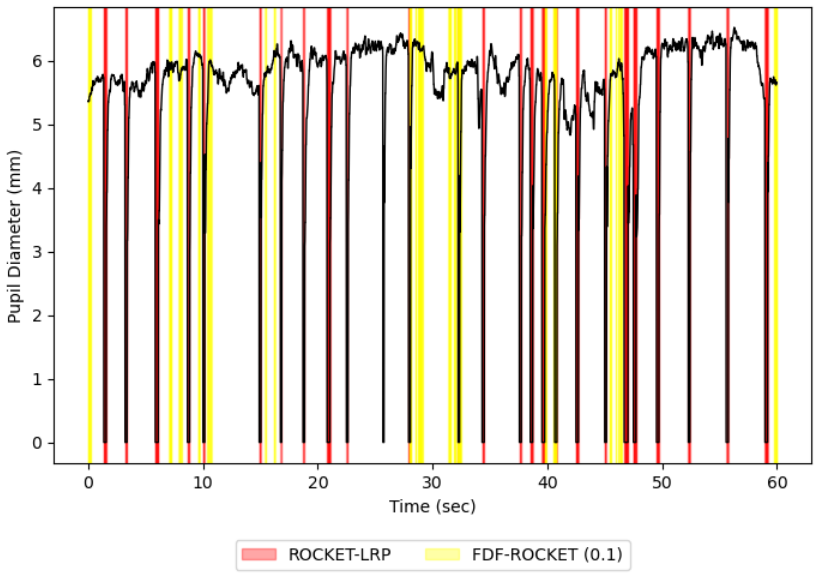


Figure 4: Top 5% most relevant points highlighted for a 60-second sample of pupil diameter data using ROCKET-LRP and FDF-ROCKET with an $\alpha = 0.1$ false discovery rate filter.

5 Discussion

Our method, FDF-ROCKET, provides a novel, statistically-informed explainability approach for ROCKET classifiers by integrating kernel filtering via feature-specific inference using empirical Bayes methods of false discovery rate estimation and activation propagation to identify influential time steps in time-series classification applications. In both simulation experiments using synthetic data and a real-world application involving the detection of alcohol-impaired driving using pupil diameter data, FDF-ROCKET exhibits superior performance over alternative explainability approaches. In particular, our simulations showed a greater ability to recover both highly localized signals and extended temporal changes.

We believe that the performance of FDF-ROCKET can in part be explained by its alignment with the unique structure of ROCKET. Unlike standard machine learning approaches, including convolutional neural networks, most of the weights used by ROCKET are not expected to encode meaningful predictive information. Consequently, explanation methods that consider all of the randomly generated kernels are likely to propagate substantial amounts of noise when determining relevance. By applying kernel selection under the framework of multiple-testing to retain only kernels with sufficient statistical evidence of relevance, FDF-ROCKET has the potential to lessen the impacts of

non-informative kernels while retaining those which reliably encode signal.

Beyond the advantages of FDF-ROCKET we demonstrate in our empirical evaluations, our analysis of the alcohol-impaired driving data highlights the practical utility of the approach. According to recent reports by the National Highway Traffic Safety Administration (NHTSA), there are currently over 100 different driver monitoring technologies under development [18]. These technologies seek to balance the high accuracy achieved by state-of-the-art time series classification methods, such as ROCKET, with explainable decision making. While both ROCKET-LRP and FDF-ROCKET identify the importance of blink-related events, FDF-ROCKET also highlights additional periods of reduced pupil diameter that are not exclusively driven by blinks. This information could potentially inform the development of classification models used by driver monitoring systems.

In terms of limitations, our approach is contingent on using an L1-penalized generalized linear model as the final classifier, as false discovery rate inference is not widely available for other types of classifiers. Additionally, the local false discovery rate estimates we use in our filtering step rely on asymptotic approximations whose accuracy might deteriorate under small sample sizes or when regularity conditions are violated. We also acknowledge that time-steps identified by our method do not inherently have a mechanistic relationship with the outcome being predicted, but instead they represent temporal locations that are associated with model predictions. Thus, the identified points should be reconciled with expert understanding of the application. Finally, FDF-ROCKET focuses on identifying time-steps of interest irrespective of the input channel. Future work may extend the ideas behind FDF-ROCKET to explicitly separate the relative contributions of different input channels in multivariate time-series data.

5.1 Conclusion

FDF-ROCKET provides a statistically informed framework for providing explanations of ROCKET models used in time-series classification by combining feature-specific inference using empirical Bayes false discovery rate estimation for L1-penalized regression and the activation propagation of a restricted set of selected kernels from the ROCKET transformation. Through simulation studies and an alcohol-impaired driving case study, the method demonstrated strong performance relative to existing explainability approaches. Overall, our method suggests the thoughtful use of model-specific statistical inference can improve the effectiveness of explainability approaches for time-series classification.

6 Acknowledgments

6.1 Data Availability

Collection of the data used in our case study was supported by funding from the *Insurance Institute for Highway Safety*. These were obtained under a third party under a data-sharing agreement and are not publicly shareable.

6.2 Generative Artificial Intelligence (AI)

The authors disclose the use of Microsoft Copilot (GPT-5) to identify typographical errors and assist in language editing. All such revisions were implemented manually by the authors, and all scientific content, analyses, interpretations, and conclusions have been verified by the authors, who assume full responsibility for the final manuscript.

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